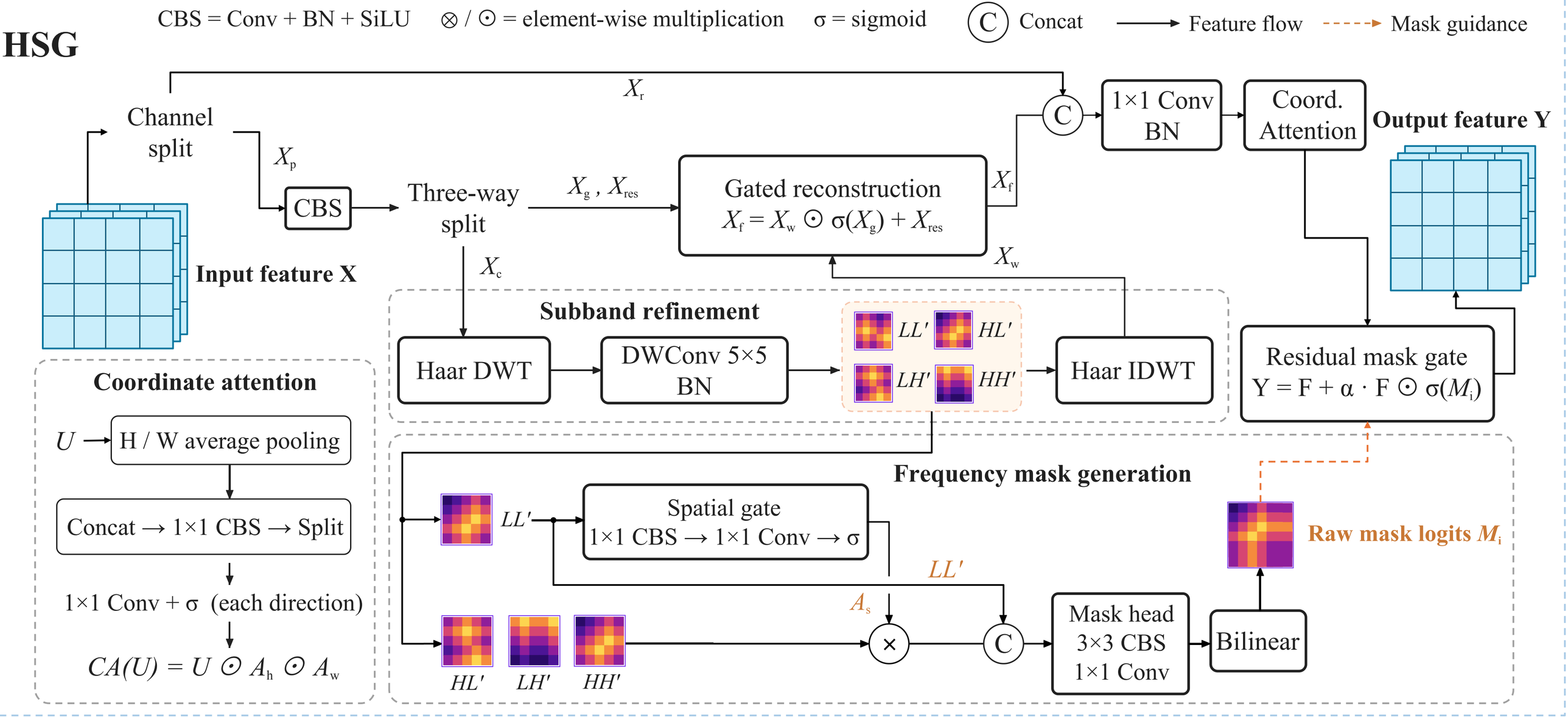

HSG
CBS = Conv + BN + SiLU
⊗ / ⊙ = element-wise multiplication
σ = sigmoid
C Concat
Feature flow
Mask guidance
X_r
Channel split
X_p
CBS
Three-way split
X_g , X_{res}
Gated reconstruction
$X_f = X_w \odot \sigma(X_g) + X_{res}$
X_f
X_w
X_c
C
1×1 Conv BN
Coord. Attention
Output feature Y
Input feature X
Subband refinement
Haar DWT
DWConv 5×5 BN
LL′
HL′
LH′
HH′
Haar IDWT
Residual mask gate
Y = F + α · F ⊙ σ(M_i)
Coordinate attention
U
H / W average pooling
Concat → 1×1 CBS → Split
1×1 Conv + σ (each direction)
$CA(U) = U \odot A_h \odot A_w$
Frequency mask generation
LL′
Spatial gate
1×1 CBS → 1×1 Conv → σ
LL′
A_s
×
C
Mask head
3×3 CBS
1×1 Conv
Bilinear
Raw mask logits M_i
HL′
LH′
HH′